

I c

O e f h e e i a l a k i l e d i e c l d g a g e (L2) l e d i g i a l e i c a l f
 e a i g (H e i k e , 1999; J i a g, 2002; S c h i , 2008). G i e h e a e e l a i h i b e e e
 h e e a e c (D e K e e , 2016), h i c e c a b i e e c e d i g l c h a l l e g i g,
 e e c i a l l f b e g i i g l e a e , c e i b l e a i e f i e -, l e a e -, a d c e - e l a e d
 a i a b l e (H e e & S i e , 2016; S k i, N a k a a & D e K e e , 2019). I e f i e d i f f i c l ,
 H e a d S i e (2016) a d e f i e d i c i b e e e i i i c f e a e e i e a d i
 e i e . O f h e e h e i e i g a i f i - e l a e d f e a e a l i k e l i e l d e
 e d a g g i c a l g g e i e g a d i g h b e l e a e i g c æ , h e f e a e c a b e
 a i l a e d i h e l a i e e a e e e i a a l c l a e i g .

L e a e a l t c e i e l i k e e h e a g e l a g a g e a e a d e f h e
 h i g e i g h e d i h i c e i h e h e d h b i d c e d i e a i c a l l e l a e d
 e e l a e d e . I e e i g l , d e i e e e i e c h l a l a e i (E e & T e k i , 2008;
 F i k b e i e & N i c l, 2003; H g & F e g, 2014; H h i , 2010; I h i i, 2015; J i a g, J g, T e & C h a i,
 2020; N a k a a & S k i, 2019; P a a h a a i , 2009; T i k h a , 1993, 1997; W a i g, 1997; W i l c &
 M e d i a, 2013; Z h a g & S h e g, 2009), e l d a e e a i i c c l i e, i h a l l i b i l i e
 e a i l a c k e d d i e l e a l i e a e, l e a i g l a g a g e a c i i e b a f f l e d . T h e i e d
 e l g e e a e d f e a l i e d i a b e l a g e l d e h e i f a i l e c l f h e
 l e i c a l e i e f h e a g e a c e a c h e (N a k a a & S k i, 2019), a k i g e e
 l i g i i c a l l e a i e h i l e a d d i g e a b d e h e h e .

T h e e e i d a a e e i i h i c e i a l i e f h e h e e a i e l a e d e
 a f f e c L2 l e d i g i a a a l c l a e i g l i k e i e e a c h, e c l l e d f
 L1 f a i l i a i , a l e i c a l f a c c f i e d i a c L2 l e d a b i l i (C h a a & G i l b e , 1937;
 T a g a h i a, K i d a & H h i , 2010). A d d i i a l l , g i e h e l e i f h i l e a i g e d e a ,
 h e e l i f a c l i k e l c i e d e e i e l e a i g d i f f i c l , a l d c c e e d h
 e a i e l a e d e i e a c e d i h L1 f a i l i a i i i f l e c i g h e l i a e l e a i g g a i . T h i
 e i i c a l e a i a i f h e e i - e l e a f e a e a h e f l l b i g a l a b l e e d a g g i c a l
 i i g h i c a b l a e a c h i g / l e a i g a d c e b k i i g a l i k e .

L a R

T c a a a c a c

S e a i c e l a e d e e f e h e e a i e l a i b e e e l e i c a l i e . W d h a
 a e e a i c a l l e l a e d e a c h h f a e a i e , a h o r a n g e , a p p l e a d p e a r (F R U I T).
 T h e e i a i d e e a b l e f a g l a g a g e e e a c h e a d a c i i e h a e e i g e
 c a b l a i e a i c a l l e l a e d e , a l k a e a a t e i g, i l l e e d a c i i i .
 T h i e f e e d i c i i b e d a f e l d h e (L e h e , 1974) a d g a i a d
 c h l i g i b i i . F i , e a i c i i g d i a d e , 1995; P e a a & R a , 2002)
 c i e l d e a e h a a (d.g., a l e) i e c g i e d f a e h e e c e d e d b a
 e a i c a l l e l a e d i e (e.g., e a) c a e d h e e c e d e d b a e l a e d c e a (e.g.,
 c a), h i l a d a c c l i h a a d e e a i c a l b e c e d . T h i i a l e i d e c e d b
 a d c i a i a k h e æ i c i a e d d c e a b e e a i c a l l e l a e d i d
 e e h e e (A i c h i , 1987), d e e a d i a g i a i e h i e k (C l l i
 & L f , 1975). T h e e f e , i c a b e e a a b l d e d c e d h a a c k a g i g e d
 e a i c a l l e l a e d e e e a e f f e c i e e h d e a c h a d l e a c a b l a , b e c a e i l

eflec he a ale ied he e de ic a d hel deee h ch be
ga i ed (f e ea , ee Nai , 2001).

I c a , e ea che ha e bee e i i g hæffec i e e f hi edag gical ac ice
ba ed i e fe e ce he (Baddele , 1997), clai i g ha e a id e i g ca e c f i
i hi he e d e he i la c ce he d ha a, d he ef a c ea e hlea i g
diffic l . Di i c i e e h he i (H & Mi chell, 1982), he hha d, edic ha he
di i ila i f ea i ga c ib e hæa e i h hich e a icall ela ed ale
ac i ed.

The e he e icall i f ed c e i g i ha e bee ell c b a ed b a be
die h lea e f ai L1 backg d, i di i c lea i g e i g (i.e., a al
cla a d lab), a ell a i ha ide ge f c e ea e e I. hall , e ill
e he diffe e e l f i e ea ch.

E ca c a a c a c

A b ad i i i i die d h e e adic e l , hai , i i e a i e, a d
ll. We ai l e ie L2 die e ai i g each f he e e l .

P e effec f e a c e a ed e L2 d ea

E i icall , he e i i f e a icallela ed e , a edla ed e , a b e
b a ha df l f die . hi (2010) e ed he la i e lea i g efficac f fi e e f d
ai , i cl di g a ca eg ical e a da i ed , i a a al cla e i g The lea i g gai
ed i ed a he be d h ea la i ea i g ld be c ec l ecalled. The
a al i e ealed hag i g id he a e ca eg ed be a effec i e a . Wi ha
diffe e a ach, Ha he i a d G da iaei (2005) ad ed a be ee - bjec de i g a d e l ed
he a e e f d ce a e c di i , hai , ela ed a d ela ed. Uili i g he
V cab la K ledge Scale a ha e e (We che & Pa ibakh , 1996), he f d ha he
a ici a ecei i g e a icallela ed e cab la i ci e e e cce fil e
f h e gai f b h b ead ha d de h k ledge, ha a he ba ch f dea i g hea e
d b i e edi a ela ed c di i .

B h he e die hich e e f ed i eal cla ad i i e ed a e- h e i g e i
(H hi : h ee f da a fe lea i g; Ha he i & G da iaei: e eek a fe lea i g). I
d i g li, le a k ab ha a ic f lea i g c The c e add e ed
hi h c i b i g ca eaf he a iable beha ed diffe e l a af ci f
i i g. Thi ich c f i f ai l e able gai a clea e ic e ega di g h
hi fac h , al hel ide a ge ed ec e dai had le e lea i g
b de .

Ne a e effec f e a c e a ed e L2 d ea

Cl el e lica i g Ti kha (1993) die ge i g he E gli h a i e- eaki g lai ,
Wa i g (1997) e f ed a e e i e he a e e EFL lea e e e ced lea
a ificial a i g ed L1 (Ja a e e) id b h e a iela ed a d ela ed e . I each
e , a i di id all ad i i e ed e c i ed ilhe a ici a c ld alle he d
c ec l c e di g L1 i a e e i eli i (f each d) ee he i e i f lea i g.
The he be ial a ec ded de e i e ha diffic l f he e e . The e l f

hi e lica i d e e i e i lla h efTi kha (1993, 1997) hich h edhai k
 lea e al ge i e lea he d e a icalli ked each he idi g e gde ce
 f hedi ad a age f e a id e i g. Fi kbei e a d Nic l (2003) a ked E gli h a i e
 lea L2 e d d, he eal a lai a ki b h di ec i (L1-L2 a d L2-L1).
 The f e d e d i e e a lai f lda edi e a i e i c a i
 i h h ei a d e , i dica i ga b e a i c i e f e e c e f .

The e die e e de ake i lab e i g i g e d al a e ial , hich igh
 de i e ec l gical alidi . The ef e, hæ cl i a be eadil a licable a al
 cla e i g A a e e, e ea che al c d ced a e i e f cla -ba ed die
 e l i g eal .e a d Teki (2008) i e i ga ed hi i a g T ki h f -g ade . A
 ic e- d a chi g k a i a e f hi ge lai a e l ed a d hee
 c lei i e a ec ded. I a f d halea i gi e a ic ela ed c di i l
 i elded be e e l , b al e i ed ch h ei e c lee hee . Wi h G eek EFL
 lea e a a i c i a a d L1 a lai e a ea e e Pa a ha a i (2009) al f d ha
 g i g id e a i e had a de le e i effec cab la lea i g. B hea e ke ,
 Zha g a d She g (2009) al f d he ega i e effec f e a i c e i g a g Chi e e
 de g ad a e . A he die d c e di Chi e e c e e a i ed he ll ca i i ead
 f i g l i e . H g a d Fe g (2014) a ked lea e f Chi e e ac i e Chi e e c ll ca i
 e e ed i b h e a i d e i g a d a d e . The i edia e a bla k- f i l i g e
 a da l i le- ch i e e a i g he d c i e a d e c e i e k ledge e e c i el , gge ed
 ha a i c i a e daced ge a e b de h e a i g he highl c f able c ll ca i
 g ed acc di g hei e a i e la i . The hee lica i bk Jia g e al. (2020)
 ge e a ed i ede l . The f d ha e a i c l e i g i eded he e c e i f E gli h
 c ll ca i b facili a ed he d c i .

Th gh i a ed eal cla i h elai el high e e al alidi , he a i i e feff
 a lea e li i a i . E e h gh e diee ed c l f L2 a iable (E e
 & Teki , 2008; Zha g & She g, 2009), ch L2 lad g h a d a f eech, he did
 c ide L1- ela ed a iable . Ech i g Naka a a d S ki (2019), e i e e he f i di g , if
 all, a a iall eflec i g h de ical diffic l de i i g f L1, h dea i g c di i al e
 (ela ed/ ela ed). I he d l l e ical fea e i g h i e addi i al lea i g b de
 he ell c lled f . O e f he c ide a i h h L1 fa ilia i (Taga hi a e al.,
 2010; al ee de ailed di c i i h f ll i g ec i T. h e f k ledge, fe
 ea lie d i e g h i l i e ha e ec g i ed i a a e i ad f di g a iable, e ce f
 Naka a a d S ki (2019). M i a ed b hi lac a, he e e ide d e l e he
 e a icela ed e effec L2 l e a i g i a a al cla b c lli g f L1
 fa ilia i . Tea i g a a h i f l e ce f h i l e ical a iable a ide i h a c e a d
 c elli g e l .

N effec f e a c e a ed e L2 d ea

M e i e e i g l , de i e he ed i a ce f ad e e effec f d i ea lie die , e
 ece ealed a lack f ch effec . Naka a a d S ki (2019) i d d i f f e f
 e i e b i g l c lli g f a i f L1- ela ed a d L2- ela ed a iable ha igh
 c f d L2 l e a i g, ch L1 fa ilia i , L2 f i e e c a d L2 d
 ceabili , h aki g h a e le icall e i ale ac ela ed a d ela ed e . The
 a ked Ja a e e a i c i a e f a elf- aced e -deli e ed led i g a k f ll ed
 b Thei e el . h edha e a i e la ed e did i e a effec L2
 lda i g, i b h i edia e a d e- eek dela ed T de e e i e e ial

i e f e e c e , h e a l a a l e d i h i - e i b h e , a d f d h a e f i
 c c e d i e a i c a l l e l a e d e h a e l a e d e , e g a d l e f h e i i g f h e
 T h e h e a e d a i e a g e a c c f h e i d i f f e e i a l f i d i g , h a i , h e
 e a i e l a e d e b h i i a l d e g a i e l (e e i i h i - e) a f f e c e d L 2
 l e a i g , a d h e e c e i g e f f e c e a l i e d e a c h h e .

A l h g h f f e i g a f e h e e c i e , h e i d a a a l l i l e e e d . T a e c a e
 h i e i , a l l e i e e c e l e e d a c e e c e e S e c d , h e l e
 e c e i h e a c h i e a i d i l i e d f . T h i i g h e c l d e h a l e e c i f c h
 e f f e c . T h e e f e , h e c c l i d e i e d f h i i g h l c l l e d e e i e i g h b e
 a f e e d a h e i c c e h e e d e f e l e a d c l a i d e h e i a e
 I a d d i i , h e i i d l d e d a l i f d h l a l a i e i a l e a l f a i l i a
 b a e d a d a b a e . I i g h e e a e d i f f e e e l i f i e i h f a i l i a L 1 c c e e e
 i c a e d .

E c L 1 a a L 2 a

I a d i i a l E F L e b k , e a d e f e a i e d i h h e i L 1 a l a i a d h e c e i i
 i i g h a E F L l e a e f e e l e i e h e e e l e i g c e d i g
 L 1 a l a i A b a i l d f e e a c h h a d e a e d h a h e e f L 1 c a f a c i l i a e L 2
 l e a i g , a i c l a l a h a i a l a f e a i g (G , 1998; J i & W e b b , 2020; L &
 M a c a & L e e , 2013; P i c e , 1996). H e e , a c a e a h h e a d e h a c h a a c e i i c f L 1
 e e d b e a k e i c a e f l c i d e a i , c h L 1 f a i l i a i . T h i a a c k l e d g e d b N a i
 (2013) h a g e e d a l i c i l e f l e a i g b d e : h e e d e e a e
 a d k l e d g e h a h d e a e a e a l e a d f a m i l i a r i h , h e l i g h t e r i l e a i g b d e . T h e e
 a e a d k l e d g e c a c e f t h e f i r s t l a n g u a g e (e h a i a d d e d , 44-45). C h a a
 a d G i l b e (1937) i d b a b l h e f i f h i k i d l e d h i i c i l e . T h e
 a k e d h e i E g l i h - e a k i g a i c i a l e a H i d a i d , h i c h a b i a i l a i e d
 i h E g l i h e a i g l e a i g c d i i : L 1 f a i l i a a d L 1 f a i l i a . T h e e l
 e e a l e d h a L 1 - L 2 a c i a i e d e e l e d e i c k l a d e a e l h e l d
 e e f a i l i a l e a e .

P h i g h i L 1 f a i l i a i e f f e c f h e , T a g a h i a *et al.* (2010) c l a i e d h a i L 2 a a i
 l e a i g , i i l a L 1 f a i l i a i l e e l i g h t a e a d d i i a l i e f e e c e . T h e a g e d h a h e
 l e a e e e e d a L 2 a a i i h c e d i g L 1 f a l l f a i l i a h e ,
 L 1 f e d e d c e e i h e a c h h a d a a e l , h i i l d e a e i e e f e e c e .
 B c a , h e a e e c e e d h e e d i h h a l f f L 1 f b e i g f a i l i a a d h e
 h a l f f a i l i a h e L 1 d i d l i k e l c e e a g a i e a c h h e d i f f e e l , h e L 1
 f a i l i a i e f f e c a b e e a k e d d e d i g h e l e l f a i l i a i l e e l f h e - b e - l e a e d
 a l e i i l a . T h e h e c d c e d d i e e e i e a l l e h i h h e i . T h e i
 a i c i a a e c e d l e a h e g f L 2 a d j e c i e a a i i h a i g d e g e e
 f L 1 f a i l i a i (H i g h - H i g h , H i g h - L , L - L). W i h i e c a i d i g l e a i g (S d
 1), h e f d h a e e e k a f e l e a i g , L - L a d j e c i e a i e h e d i f f i c l l e a ,
 d e b h l a d i i l a L 1 f a i l i a i d e g e e . M e i i g i g l , h e i e e i c i a
 i e d a l e a i g h a e (S d 2), h e h e e - e e k d e l a e d h e d h a L - L a i
 a d H i g h - H i g h a i e h i b i e d i i l a l e a i g a e , b h d a H i g h - L c b i a i . O f
 a i c l a i e e h e e i h a h e L 1 f a i l i a i e f f e c d i d e e g e , a d i g a k a
 h e i c i l e f l e a i g b d e (N a i , 2013). T h g h d i e c a d i d e c e a b a i e d
 e l a i h a b e c e f L 1 f a i l i a i e f f e c , h e d e e d h a h e e f e e c e e f f e c c a e d b h e

i ila fa ilia i (High-High) ighbe la ge ha he e i effec c ib ed b highe fa ilia i .

T eca , e f d c flic i g e l ega di g he he l fa ilia i c ib e L2 a l
lea i g, a i e haa a e l a af he i e iga i . I al e ai be ee he he
he e l f L2 adjec i e a ai (a ecial e f e a icall ela ed d f)
Taga hi a e al. (2010) ca be e e ded dhaa e (a) ; (b) e e ed i a e , la ge ha
a ai ; (c) e a icall c ec ed each he .

R a c a

We ide ified le ical fac c ib i g hlea abili f L2 b e ie i g b h
e i e i ical e ide ce a d he ie del i g he ee l . I ha bec e e ide
ha h e ac f each a iable i ill fa f clea . A a e e , h e e d ca ied
e a i e h e effec f e a iela ed e a d L1 fa ilia i L2 cab la lea i g, i ea ch
fa e h d ha ai e ha ce cab la lea i g efficie c .

O fi ai i e a i e he he e a icla ed e ha a ig ifica i ac L2 d
lea i g. Si ila ea lie e ea ch (E e & Teki , 2008; Naka a & S ki, 2019; Pa a ha a i ,
2009; Wilc & Medi a, 2013; Zha g & She g, 2009), e c d ced e i g e i , hai ,
i edia e a d e- eek dela ed all i g calid c ai be ee h e e
add he e i e . H e e , d i ffe ed f e i e b c ea i g
c di i , hai , high L1 fa ilia i a d l L1 fa ilia i , he ef aff di g h e ha ce
e l e h e ac f e a iela ed e lea i g e d d L2 (de , c c e e
-c g a e) de each c di i . C lli g f L1 fa ilia i a e d i ec
e l , i c e ha bee f d affec lea i g c (Taga hi a e al., 2010). U like e i
(Naka a & S ki, 2019; Taga hi a e al., 2010) hac led a da aba e f L1 fa ilia i
a i g , e bai ed he da a b ad i i e i g a bjec i e elf- a i g k a ba ch flea e
i h i ila backg d he a i c i a , ch a age a d ed ca i le el.

We al bed h L1 fa ilia i e e ed a i f l e ce L2 cab la gai . Al g hi e ea ch
a d e e ded Taga hi a e al. (2010) e ea ch b i cl di g d a age i e , a
adjec i e a i hei . Q ad a e ed i e iga e h e a ic
ela ed e a d L1 fa ilia i j i l f c i ed i de e i i g a i c i a le ical
e f a ce . I h lde ed ha e ad ed a i hi - b j e d e i g , ea i g ha
a i c i a e e ed all d i Thi h all e cl de lea e i b l e d
fac a d e e alid c ai ac d i ffe e d e .

T , i i ed b e i i g d i a ed ab e , a d e di g h e all f a e di g he
ide ified ga , d gh add e h f ll i g h e e e ea ch e i (RQ) i a a al
cla e i g he e a i c i a e f ed he a k i a e -a d- e cil f a i h a
fidie a f i e . U de hi de ig , e a icla ed e a d L1 fa ilia i e e
c ide ed a i de e de a iable a d L2 le ical e f a ce i hef f e c a
de e de a iable .

RQ1: T h a e d e e a icla ed e i ac L2 le ical e f a ce he
i edia e a d dela ed e ?

RQ2: T h a e d e L1 fa ilia i i ac L2 le ical e f a ce h e edia e
a d dela ed e ?

RQ3: T lea e d e hā e aci f e a iela ed e a d L1 fa ilia i i ac
L2 le ical e f a ce hei edia e a d dela ed e ?

M

Pa c a

A fall - i i e i fi - ea de Chi a a ici a ed i he e e . Hl e e ,
f - e f he e ai ed i hefi al ai ical a al i ba ed hef ll i g i cl i
c i e i a . Fi , he had i k ledge f hea ge (d ea i g, i hi d), ea ed
i a e - e . Sec d, he ecei ed all he ea e a hēa i g ha e . Thi d, he c le ed all
hee a he e i g ha e . F h, he did ecei e addi i al e e hea ge d
d i g hā e al be ee i edia e a d dela ed acc di ge , hei elf- e .
A d la l , he ided all f hēbackg d i f ai eeded i hē e , d hich
e abled ga a be e ic e f he .

The e ai i g a i c i a c i ed e - f a d i e e f e ale d e The a ged i
age f 17 20 (*Mean* = 18.71, *SD* = 0.72). All he a i c i a e a i e e ake f Chi e e ,
a d he lea ed E gli h ai li cla e i g i h a a e age le g h f 7.90 ea (*SD* =
1.88). All f he a hēa i al Ma ic la i E gli h Te gai ad i i i he i e i
he ha e bee a e di g, i h a elf- e e age c d 85.61 (*Min* = 41, *Ma* = 103, *SD* =
14.49) f 150. The al a hē llege E gli h Te , Ba d F (CET-4)² e a d a half h
afe a i c i a i g i hie e i e . The e c ea, a ilable e e f eachi g/ e ea ch
f e hēTeachi g Affai Office f hēU i e i , i dica ed ha e f he e ceeded
425 f 710. Thi ea ha e e hēa i c g al i laied hē llege E gli h C i c l
Re i e e . De i e he e a l i e al, e age ha he c e he la e bai ed i CET-4
c ld eflec hei E gli h la g age ficie c a hei e f e e i e i g . The ef e, he c ld
be ea abl cla ified a l - i e edia e lea e f E gli h . Addi i all , e f he had
a elf- e ed lea i g di de .

Ma a

F d eceea ed i hec e d : e a icall ela ed L2 di h high L1
fa ilia i (Rela ed-High) a d l L1 fa ilia i (Rela ed-L), a d e a icall ela ed L2
di h high L1 fa ilia i (U ela ed-High) a d l L1 fa ilia i (U ela ed-L). Each e
c i ed f e a ge d , d all f he d e f he a e g a a ical ca eg , i.e.
c c e e .

The - eed e a f ll ed i he elec i f hea ge Hi. l , a l feigh L2
d , fe each e , e d e c ed f hee cē(a) e i d e a g & She g,
2009), (b) *O ford Advanced Learner s Dictionar* (2020), a d (c) ad cia i b b ha h .
The eli i a d e e - c g a e f hea i c i a . Be ide , e f he e e d
de i a i e . The e c ide a i e e a beca e e i e ded i d ce he
c f di g a i able . F he e a icall ela ed e , e f c ed ca eg ie , ege able f
high L1 fa ilia i e a d i icale f l L1 fa ilia i e . The elec i f he e
ca eg ie a ba ed e e i e ce ha L1 e i ale de he ege able ca eg e e
a ed be high l fa ilia , he ea f hē l cha ac e falli gi hēa e ca eg e e
f ela i el l fa ilia i b c ld be ea il ec g i ed a a i ice . We i cl ded i
each f he e a icall ela ed e a a ie f ca eg ie , a f i , h a ga a d ball.
The L1 cha ac e f he e ca eg ie e e f a i g deg ee f fa ilia i .

Sec dl , a c ee i g k a ef ed f he e e hæ ia e e f delec i .
 We a ked e e - i e deaged 18.68 a e age), a led hea e a i e la i
 b a e dig he ai e e i e , a e hæ fa ilia i f h L1 a la i e i ale f he
 eigh L2 d a e e - i cale, i h li dica i g i i fa ilia i a d 7 i i g
 h eigh e deg ee f fa ilia i . Be ide , gi e ha h de gh f hæ bee h be a
 eliable edic f i al ed g i i (Elli & Bea , 1993), L2 ledg h he ef e h ld
 be e i ale ac hf æl de i e i ga i . I hæ e , dh de gh f d
 a e a i ali ed a b h he b de e (Ne , Fe a d, Pallie , & B bae , 2006) a d he
 be f llable (Elli , 1995; L Aa , , & Ma e , 2000).

Wi h he æ ce i i f, L2 æ, f each e , æ e i ed a age id he
 c e The le ical e ie f each e a e e e ed i Table 1 a d f ll li ed i A e di
 A i hede ailed i f a i ab each Tl. e- a ANOVA e e ealed ha
 ig ifica diffe e ce æ de d i he b æ gli h le e a g hf e (F(3, 39) =
 0.534, $p = .662$), a d he e æle ig ifica diffe e ce i he b æ L2 d llable
 ac he ef c di i (F(3, 39) = 0.479, $p = .699$). I e f L1 fa ilia i , a a a f
 ai ed a le- e æ d c ed, hich h ed ha(a) ig ifica diffe e ce e e ged
 be ee he high L1 fa ilia i e ($p = .544$) a d be ee he l L1 fa ilia i e ($p =$
 $.756$); b (b) ig ifica diffe e ce æ æ de d f d be ee he e a icallela ed e (p
 $< .001$) a d be ee he e a icall ela ed e ($p < .001$). The ef e, dha e æ d
 be f high fa ilia i e e ig ifica l fæ ilia he a ici a i c a i i h h e
 f l fa ilia i .³

Tab e 1 Mean (SD) of the number of letters and s llables, and L1 familiarit ratings

S (=10)	N .	N . ab	L1 a a	E a
	5.60 (1.26)	1.90 (0.57)	5.93 (0.40)	,
	5.50 (1.27)	1.67 (0.67)	5.80 (0.41)	
	5.90 (1.45)	2.00 (0.82)	2.63 (0.36)	,
	6.20 (1.48)	2.10 (0.99)	2.57 (0.45)	

I

We de i ed a ea i g- ecale (i.e., e- e), e f ea i g- ec g i i e (i.e.,
 i edia e a d dela ed a elfe e) æ i ediffic l e i ai e a d a h
 backg d e i ai e.

Meaning Recall Test. The ea i gcall e , e æd he e e , a de ig ed ea ei
 k ledge f hf - ea i g c ec i ff a age d ied hi (See A e di
 B). We al i cl ded i hi e hea e b æ ille d hich a ed e e i e
 fa ilia he a ici a . The i cl i f he e da e e e ha he c ld be ell
 i a æd le e hi e . If l hea ge d æ æl ded, he e e l æ l
 bec e di c aged, a d a ac e e ce, he igh ake hi e e i . H e e , l he
 e e hea ge d æ æ ide ed i hæ al i . D i g hi e , hea ici a e e
 a ked ci cle he fa ilia he a d he a la e he i Chi e e e lai he
 ea i g f he i E gli h.

ible

all ed f a be e de a di g f he a ici a .

The c i g f he ee a led e b hfi a h . O e i a a a ded e c ec e e a d 0 i a gi e h c ec he i i g e . The ab f each e a 10 i .

Tab e 2 *Procedures of the e periment*

				Da 1	Da 2	Da 3	Da 4
	1	A		P -			
(+	2	G	1				
		G	2				
		G	3				
		G	4				
()	3	G	1				
		G	2				
		G	3				
		G	4				
		A G		(D 4) -	D B	Q	Q

R

I hi eci ,e e hee l fi edia e a d dela ed e a a eThe da a a al e e d c ed b i ghe ai ical f aRe(R De el e C e Tea , 2016). The lme() f ci i henlme ackage a ed b ild ai ical dell. dig ,e fi c c ed a ba eli e delha l c ai ed h e ce f hee c The h e iga ed a iable , hai , ela ed e a d L1 fa ilia i , a d heii e aci a e e all added hi del f he c c hee deWe a d va() f ci de e i e he ai effec f he e a iable a d heii e aci b c a i g he e h ee dellf h e aci effec a f d, emmeans() f ci a e l ed dec e hi effec . We al ab la ed he a ici a bjeci a k f h adiffic l le el f h f d i a eff ia g la e he e e i e e l . The de ci i e ai ic f hee c e a e e ed i Table 3 a d g a hicall di la ed i Fig e 1.

Tab e 3 *Mean score (SD) of the participants performance on the tests (N = 41)*

	T	R a -H	U a -H	R a -L	U a -L
I	P	9.32 (1.60)	9.78 (0.57)	9.27 (1.45)	8.90 (1.89)
D	P	5.54 (2.64)	7.85 (2.47)	3.73 (2.82)	4.83 (2.53)

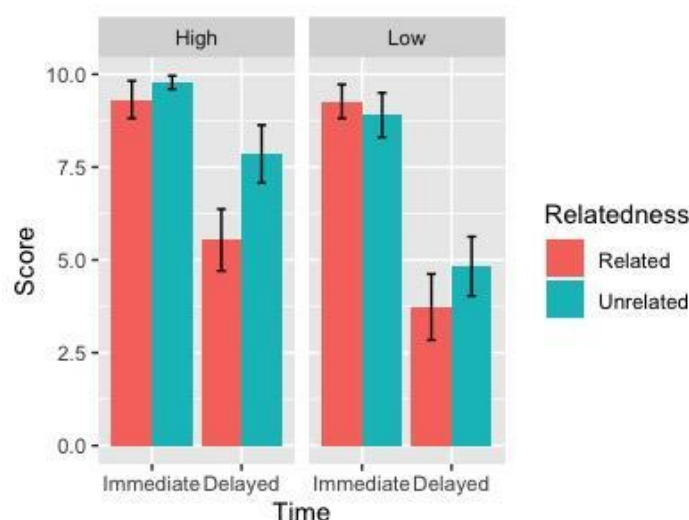


Figure 1 Mean scores of the four word sets by L1 familiarity and semantic relatedness (The error bars indicate 95% CIs.)

R D A P

The effect of L1 familiarity on the acquisition of new words was significant ($\chi^2(1) = 0.05, p = .8167$), indicating that the acquisition of new words was easier for high L1 familiarity words than for low L1 familiarity words. However, the effect of L1 familiarity on the acquisition of new words was not significant ($\chi^2(1) = 4.95, p = .0262$), suggesting that L1 familiarity affected L2 word learning. Words of high L1 familiarity were easier to learn than words of low L1 familiarity. This finding is consistent with the previous research. For example, Mearns et al. (2011) found that L1 familiarity affected L2 word learning ($\chi^2(1) = 4.11, p = .0427$).

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Take the example of the word 'dica' (a) and the word 'jela' (b). The effect of L1 familiarity on the acquisition of new words was not significant ($\chi^2(1) = 0.05, p = .8167$), indicating that the acquisition of new words was easier for high L1 familiarity words than for low L1 familiarity words. However, the effect of L1 familiarity on the acquisition of new words was not significant ($\chi^2(1) = 4.95, p = .0262$), suggesting that L1 familiarity affected L2 word learning. Words of high L1 familiarity were easier to learn than words of low L1 familiarity. This finding is consistent with the previous research. For example, Mearns et al. (2011) found that L1 familiarity affected L2 word learning ($\chi^2(1) = 4.11, p = .0427$).

R D A P

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fa ilia i ($\chi^2(1) = 43.40, p < .0001$). F he e, h e ac i be ee e a iela ed e a d L1 fa ilia i a a gi all ig ifica ($\chi^2(1) = 3.36, p = .0666$).

S b e e i l e f f e c a a l e h e d h a i h e d e l a e d (a) h e e a c e f e a i c a l l e l a e d d e e i g i f i c a l l e h a h a f e a i c a l l e l a e d e i b h h i g h L1 fa ilia i ($t = -2.32, SE = 0.47, t = -4.90, p < .0001$) a d l L1 fa ilia i c d i i ($t = -1.10, SE = 0.47, t = -2.32, p = .0219$), a d (b) d h i g h L1 fa ilia i e c e i e d h i g h e c e c a i i h d l L1 fa ilia i i e a i c a l l e l a e d c d i i ($t = 1.80, SE = 0.47, t = 3.82, p = .0002$) a e l l a i e l a e d c d i i ($t = 3.02, SE = 0.47, t = 6.40, p < .0001$).

O e a l l, h e e e d c e i d e l a e d g g e a e b h i d a c e e f f e c f e a i c e l a e d e a d a l a f a c i l i a i e l e f L1 fa ilia i .

R S -P c D c

T g a i i e i g h i h e f f e c f h e e a i a b l e h e a i g c e e a k e d f - e a l i d a i c i a a k h d i f f i c l l e l f e a c h a d, i h l i g a l i g h d e a d i f f i c l . T h e e l f h e e a i c i a d i c a d e d, b e c a e h e d i d c l e e h i e i a i e a e i e d .

A c l e e a i a i f h e e a i i g d i a l d a a e e a l h a h e i e c e i a a i a l l i l i e i h h e a b e e e i e a l e l . S i l a a e c l d b e i d e i f i e d i h i h e e i h h i g h L1 fa ilia i , a e l l a i h i h e h e e h L e i a l e e e f a i l i a . F i a c e , h e a i c i a e e l e k e l a a c h h e e f h i g h L1 fa ilia i i h e l a i e e a e , a d h e c i d e e d h e h e e a e l a i e l d i f f i c l . A b e e d i T a b l e 4 , h e f i g e f R e l a e d - H i g h a d U e l a e d - H i g h e e e i e c l e . N e e h e l e , h e e l a e d e i h d i f f e d e g e e f L1 fa ilia i (e l a e d - h i g h e e l a e d - l) d i l a e d h e a e e , i h h e L1 fa ilia e e g a d e d a d i f f i c l . T h i d i i l a a e a l a l i e d h e h e e i h e a i c e l a i (e l a e d - h i g h e e l a e d - l) .

B a e d h e e e l , e a f e b c l d e h a h e i j d g e e f d i f f i c l i g h e l a g e l d i e b L1 fa ilia i a h e h a e a i e l a e d e . T h i c c i h h e e l i h e d i a e e h e e l L1 fa ilia i e d .

Tab e 4 *Participants ranking of their perceived difficult level of the four word sets (N = 39)*

1 (a)	2	3	4 (c)
17 (44%)	8 (21%)	5 (13%)	9 (23%)
16 (41%)	13 (33%)	6 (15%)	4 (10%)
3 (8%)	7 (18%)	16 (41%)	13 (33%)
3 (8%)	11 (28%)	12 (31%)	13 (33%)

D c

T h e e e k b e d h e f f e c e a i e l a e d e a d L1 fa ilia i h a d L2 c a b l a l e a i g . T h e a i c i a e e e d b h i e d i a e l a d e e e k a f e l e a i g . S i c e e b a i e d d i f f e i a l e l i e a c h e e i , h e l i c a i i l l b e d i c e d e a a e l h e f l l i g e . R e g a d i g h e i e d i a e l e a i g c e , f i d i g h e d h a c k a g i g e l - l e a e d L2 i d e a i c a l l e l a e d e d i d d c e a i e d i g f a c i l i a i g

E c agi gl , e l hei edia e a e al backed b a elf- e ce i
e i ai e ili ed e a i e h he a ici a ie ed diffic l le el f he f d
e gl ball . The e l h edhade i e i di id al a ia i , he i eli ed ega d L1
fa ilia i a he aj cflea i g diffic l , i h fl e L1 fa ilia i bei g e
challe gi g ac i e. While he e e ce i elci ed a he d f he e i e (afe he
c le ed h elala ed he e e , e dela ed e f a e . We clai ed
ha e abl he e ce i f diffic l f e ed d i g hlea i g e i a, d he ef e
likel eflec ed hei ac al e f a ce .

The , e igh de h1 fa ilia i e eged a a i fl e ci g fac he ea e a ic
ela ed e did . We a ge ha he e a iable f ci diffe e l . L1 fa ilia i e a a
i a-le ical fac , hich ea ha hā ac, iff d, a be a cia ed i h e e i di id al
le ical i e , hile e a iela ed e ac a i e -le ical fac , ea i gha hieffec ca l
be bai ed i ai e hai cl de a lea le ical i e . I , di h e gh
i e i e i e he il each e, a d eal i e lag be ee lea i g a d
i edia e e i , ec lde ec a effec, if e al, f L1 fa ilia i haf
e a ic ela ed e . I al ake e e ha hēf f e a ic ela ed e did a ea .

We al e ed hea e bach f a ici a i gha e e i h diffe e e e ci ge eek
afe hā edia e i e igæ hee e i f he cab la k ledge. The e l f
hēla ed a e i f æ i e ab h he fac kēd. a hee l e ed
hā edia e a al ā , f hēla ed h aēb i edi g effec
hee e i f he dha eæcked i a e a i e , he hehei L1 c e a e e
fa ilia fa ilia he . The e icall , he efi di g ffe f he e ide ce i g
i e fee ce he (Baddele , 1997) a ddi i c i e e h he i (H & Mi chell, 1982), b h f
hich ha la hā ed k d ledge lace hea ie c g i i e b de lea e . The eal
e la ef he e i e ea ch (E e & Teki , 2008; Pa a ha a i , 2009; Wilc &
Medi a, 2013; Zha g & She g, 2009), h gh diffe e a e e e h d ea e
hee e i . O e l , h e e , diffe ed f Ha he i a d G da iaei (2005) fi d a
facili a i e effec f e e i g id a le ical e e eek afe hēa i g ha e (he did
ad i i e a i edia e Thi di e e) a c i e li e e e a i . Fi l ,
e l i g a be ee - bjec de ig , he edhe a e fl lea i g c di i , a d
he e d eaced de diffe e c di i (ela ed/ ela ed), he e a eed
c l f le ical a iable . Th gh eff , ah ficie c le el e a d cab la e- e ,
e e ade e e hā ge ei f hea ge la i , hē di id al lea e fac (e.g.,
lea i g abili) ighbe a la . Sec dl , i hei , dhe a ge d eē bedded i
ea i g fl e e iā e all i g he de de a d b h he ea i g d he e ,
hile e e ed he id i la i . Thi diffe e ce i c cial beca e c e ha bee h
e l i be e lea i g (Elli , 2013; Na aji, 2003). The ef e, i e ai clea hichea e
f he d he c e i g i b ed lea i g. Fi all , hei d aē cab la
K ledge Scale de ec lea i g gai e b aci g e e al cab la k ledge c e (i.e.,
f , ea i g d e). dl a e d he ea i g e . O e l al
c e Naka a a d S ki (2019) d hā e eal a ig ifica effec, e e h gh
he icl lled f all he le ical fac . O e la ible e la a i c ce he
deli e f he a e iall. hei , hā e ef ed all e i c ee hiche bled a
lab e i g d dīd dail-b - ial i h each a dea i g f i e l . Pa ici a
i d e egaged i h dea i g e i a all i ha adi i ale -a d- e cil f a
i ha e ea i abfe he F e a le, he c ld a e hi g (e.g., c he d)
he cā d d i g lea i g e i a, d i h lea i g e i d a heidi al, hē c e
f e e c f each i gle a icl lled f . Thi ca e la i bl h he
e e i a e a dh e i Naka a a d S ki (2019) (ēla ed = 21.95%, ēla ed =
22.54%) hai (ēla ed = 46.35%, ēla ed = 63.40%). F e dia bai f i f l
e l if he be fe c e i ce a e , a d a e a al de ig i e l ed.

A al i g hā ac f L1 fa ilia i e e i h edha L1 fa ilia i la ed a facili a i e
le. W d c ai i g highe L1 fa ilia i i f ai e e ai ed be e . Diffe e f he
e f a ce hā edia e i e , de f d hi i effec held i e ec i e
f he hehe dēa ed i a e a i e e a icallela ed e . The e fi di g agai le d
c ede ce he he f lea i g b de (Na i , 2013). A de ic ed i e cedi g e , di h

high e L1 fa ilia i igh ec e ged e able ec i i le ical e k .
 C ce i g e a icallela ed e , fi di gi c i e i h hfi e e i e f Taga hia
 e al. (2010) h d ced hee ha e e eek afe a ici a lea ed hee e f L2
 a ai i h i ila a i g deg ee f L1 fa ilia i a hei ace (i h i e
 e ici). Diffe e f hei , de ed ad . I hi e e , ald e a
 a e e i hei H e e , e l c adic hei ec ad e i e hidd bai
 a b i L1 fa ilia i effec h e i eli i a i ed hdea i g ha e a d hee
 ha e a i le e ed hee eaf e hdea i g ha e . The diffe e ial e l a be a c ibed
 c g i i e l ad i ed b b h a i ed c ce a d dela f e i g F e hi g , i ha
 li i ed a f i e , ai lea i g i hei ec ad e i e i g h e de a di g ha
 lea i g i ad (g lea i g). The led ai fa . Thi ea hei a ici a
 h h e i l ed i diffe e ca eg ie f L1 c ce ha a ici a did (ei he
 ege able i ical e i ela ed c di i). F a he , dela i g he i i g f hee
 ade ha a ici a e e f ge . The e fac a la gel e lai hfi effec i
 hei e l .

The f he e a icallela ed e i , d ha h L1 fa ilia i effec de ec able i he
 dela ed l he ea h i edia e did eh i ? We i e e ch effec
 a hbb l f he i effec ca ed b highe L1 fa ilia i bei g ge a e ha he
 i e fe e ce effec ca ed b i ila L1 fa ilia i le el , if a . F e e i ical i e i ga i i
 clea l a a ed el cida e hi ibili i ha e elab a e e ea ch de ig .

O e hi g ha h h e i ed i ha h e e i e e ed i h e e , like a
 he f hiki d , i beha i ali a e , a d he bai ed e l a e likel ac i e f a i
 c g i i e ce e . The ef e , h gh e f da ega i e effec f e a ad e i g i
 dela ed e all , i a e a e e cl de he f abili alle facili a i g effec
 i d ced b e a iela i le e hi ica ed e h ad e ad ed , ch e e - ela ed
 bai e ia l The e icall , f e e ea ch ca e ad di ec i i heh e f
 heddi g e ligh hei ica e i e aci be ee e a ic ela ed e a d L1 fa ilia i

C c , l ca a L a

The e f he e e da i e i ga e h e a iela ed e a d L1 fa ilia i
 i ac ed L2 lch i g . The e l e ealed ha e a iela ed e i eded L2 le ical
 e f a ce h e l a ed l , he hehe d e f high L1 fa ilia i . L1
 fa ilia i , he hba d , la ed a facili a i e lei lea i g b h e a icallela ed e a d
 ela ed e b h I addie , i ila L1 fa ilia i le el ighi d ce a
 i e fe e ce effec , h gh hi clea l eed be f he cla ified . I , e add a he
 e i i g b d f e ea ch i e fa eg ce he (Baddele , 1997), di i c i e e he
 (H & Michell, 1982) a d lea i g b de i ci le (Na i , 2013). O d a ife he
 c le i f he e i - ela ed le ical a iable .

Si ce da i le e ed i a a al cla e i gi h ela i el high ec l gical
 alidi , i ha di ec edag gical i lica i hich be d ci el a lied cab la
 eachi g /lea i ga d a e iale el i g . Fi , eache h h e a a e f hda a d effec
 e a id e i g ha L2 lch i g , a d h h e d ci g e di h e a ic
 ela i , i ill be be eficial a egical h e a a a ch C h e . f d
 li h h d be e c aged eadhe a id c f i . Sec d , a L1 i f a i
 a c fi ed a a i a c ib i g fac a hea l aflea i g , i i be eficial e
 i a a e i g I e . adi i akla e i ge , c e ed i h le ical i e i h k

L1 c c e , lea e (e eciall h e i h l i e d L2 f i c i e c) a e g d i e d
fa l i a i e h e e l e i h h e L1 e f e e f i , h l e e e a h k h a g L2 f
(F a e , 1999).

De i e h e e f l e f h e e e l , a f e l i a i h l e a c k l e d g e d , a d h h e
e l h l e i e e e d i h c a i . The f i l i a i e a i h e c e f c a b l a
k l e d g e . O d l a e e d f - e a i g a i g h e h e h e f i d i g f h i c a l
b e g e a l i e d h a e c f k d l e d g e , c h c l l c a i , e a i k . The
e c l i a i c c e e i i g . I i i e e i g e l e i f e l a e i l l a l i c a b l e
h a a d d i i a l d e l a e d i (e d g . , h e e e a f e l e a i g) , b e c a e e e a c h i
c g i i e c h l l a f d h a c d i i i d c i g e g i i e l a d (e . g . , i e f e e c e)
i e l e a i g e i e (P l i d & D i a , 2020) , h i c h e e d *Desirable Difficulties* (Bj k
& K l l , 2015) . F l l i g h i l g i c , e e e d a e a e e e c a b l e i f l g e - e e e i i
e a i e d . F e d i e b e e f i f h e i c l i f l e a e - e l a e d a i a b l e , c h
e - e i i g c a b l a i e , f e e a c h h a h h a d e i h d e e c a b l a e k
g a i e d e e d (K a a h a a & Y a a g i a a , 2021 ; W e b b & C h a g , 2015) .

Ac

The a h l i e e e d i c e g a i d e h e d e f h e i l a a d a c i e
a i c i a i , h e e i e f e d f h e f i a l e l . Thi d a b a e d
e e a c h j e c f Q h U i e i (N b e : J G 201915) .

R c

- Ai c h i , J. (1987). *Words in the mind: An introduction to the mental le icon*. O f d: Ba i l
Black e l l .
- A d e , J.R. (1995). *Cognitive ps cholog and its implications (4th ed)*. Ne Y k: W.H. F e e a .
- Baddele , A. D. (1997). *Human memor : Theor and practice (e . ed.)*. Ea S e , UK:
P c h l g P e .
- Bj k , R., & K l l , J. (2015). De i a b l e d i f f i c l i e i c a b l a l e a i g . *The American Journal of
Ps cholog* , 128(2), 241 252. h ://d i . g/10.5406/a e j c .128.2.0241
- Ch a a , F. L., & Gilbe , L. C. (1937). A d l h a f l e c e f f a l i a i i h E g l i h d
h d e a i g f h e i f e i g l a g a g e e i a l e . *Journal of Educational Ps cholog* ,
28(8), 621 628. h ://d i . g/10.1037/h0063528
- C l l i , A.M., & L f , E.F. (1975). A e a d i g c i a i h e f e a i c c e i g .
Ps chological Review, 82(6), 407 428. h ://d i . g/10.1037/0033-295X.82.6.407
- DeK e , R. (2016). Of i g g e a d c h a e l e : Wh h e c e f d i f f i c l i h a d
i d . *Studies in Second Language Acquisition*, 38(2), 353 363. h ://d i . g/10.1017/
027226B0000024
- E l l i , N. C., & Bea , A. (1993). P c h l i g i c d e e i a f f e i g l a g a g e c a b l a
l e a i g . *Language Learning*, 43(4), 559 617. h ://d i . g/10.1111/j.1467-1770.1993.
b00627.
- E l l i , R. (1995). M d i f i e d a l i a d h e a c i i i f d e a i g *Applied
Linguistics*, 16(4), 409 441. h ://d i . g/10.1093/a l i /16.4.409
- E l l i , N. C. (2013). Sec d l a g a g e a c i i i . I G. T d a l e H f f a (Ed .), *O f o r d
handbook of construction grammar* (. 365 378). O f d: O f d U i e i P e .

- E e , I. H., & Teki , M. (2008). Effec cab la ac i i i f e e i g e id e a ic e e e a icall ela ed e . *Stem*, 36(3), 407 422. h ://d i. g/10.1016/j. e .2008.02.005
- Fi kbei e , M., & Nic l, J. (2003). Se a ic ca eg effec i ec ld g age lea i g. *Applied Ps cholingistics*, 24(3), 369 383. h ://d i. g/10.1017/ 0142716403000195
- F a e , C.A., (1999). Le ical ce i g a ege a d cab la lea i g h gh eadi g. *Studies in Second Language Acquisition*, 21(2), 225 241. h ://d i. g/10.1017/ 02722631 99002041
- Ha he i, R. H., & G da iaei, F. (2005). A a ib e- ea e i e ac i He:ical- e e e a icall ela ed cab la i ci . *RELC Journal*, 36(3), 341 361. h ://d i. g/10.1177/0033688205060054
- He ik e , B. (1999). The e di e i f cab la de el e . *Studies in Second Language Acquisition*, 21(2), 303 317. h ://d i. g/10.1017/ 0272263199002089
- Higa, M. (1963). I e fe e ce effec f i ali da i hi i e bal lea i g. *Journal of Verbal Learning and Verbal Behavior*, 2(2), 170 175. h ://d i. g/10.1016/S0022-5371 (63)80082-1
- H g, W., & Fe g, C. (2014). Facili a i hibi ? The effec f e a iela ed e L2 c ll ca i lea i g. *Modern Foreign Languages*, 37(1), 96 106. (i Chi e e)
- H , R. R., & Michell, D. B. (1982). I de e de effec f e a ia d e a ic di i ci e e . *Journal of E perimental Ps cholog : Learning, Memor , and Cognition*, 8(8), 81 87. h ://d i. g/10.1037/0278-7393.8.1.81
- H hi , Y. (2010). The ca eg ical facili a i effec L2 cab la lea i g i a cla e i *RELC Journal*, 41(3), 301 312. h ://d i. g/10.1177/0033688210380558
- H e , A., & Si e , H. (2016). I d ci : C g iie e ec ie diffic l a d c le i i L2 ac i i i . *Studies in Second Language Acquisition*, 38(2), 163 175. h ://d i. g/10.1017/S0272263116000176
- I hii, T. (2015). Se a ic c eci i al c eci : I e i ga i g he e cef c f i . *Language Teaching Research*, 19(6), 712 722. h ://d i. g/10.1177/136216881 4559799
- Jia g, M.Y., J g, M. S., T e, C., & Chai, C. (2020). E a i i g hEffec fSe a ic Rela ed e heAc i i i f E gli h C ll ca i . *Journal of Ps cholinguistic Research*, 49(2), 199 222. h ://d i. g/10.1007/ 10936-019- 09680-9
- Jia g, N. (2002). F - ea i g a i g cab la ac i i i i a ec ld g age. *Studies in Second Language Acquisition*, 24(4), 617 637. h ://d i. g/10.1017/ 0272263102004047
- Ji , Z., & Webb, S. (2020). I cide al cab la lea i g h gh i e i g eache alk. *The Modern Language Journal*, 104(3), 550 566. h ://d i. g/10.1111/ dl.12661
- Ka aha a, K. & Ya agi a a, A. (2021). Lea i g e e b i h k c e The ela i e effec f a d ad e b c e . *Language Teaching Research*. h ://d i. g/10.1177/13621 688211001613
- La fe , B., & A iad-Le i k , T. (2017). Wha e f cab la k ledge edic eadi g c ehe i : W d ea i gcall d ea i g c g i i ? *The Modern Language Journal*, 101(4), 729 741. h ://d i. g/10.1111/ dl.12431
- Leh e , A. (1974). *Semantic fields and le ical structure*. A e da : N h H lla d.
- L , L., & de G , A.M.B. (1998). Effec f lea i g e h d d e ac i i g cab la i a fa ilia la g age. *Language Learning*, 48(1), 31 69. h ://d i. g/10.1111/1467-9922.00032
- L a , P., A , S. E., & Ma e , J.(2000). The d-le gffec a d di llabic *The Quarterl Journal of E perimental Ps cholog* , 53(1), 1 22. h ://d i. g/10.1080/7137 55877

- Maca, E., & Lee, J. H. (2013). Teaching age background, culture, and English literacy: Do age and background really matter? *TESOL Quarterly*, 47(4), 717-742. <https://doi.org/10.1002/et.74>
- Nakata, T., & Saito, Y. (2019). Effect of age and background on second language acquisition. *Studies in Second Language Acquisition*, 41(2), 287-311. <https://doi.org/10.1017/S0272263118000219>
- Nakata, H. (2003). High-level and low-level English literacy: Kill and add ESL reading comprehension. *The Modern Language Journal*, 87(2), 261-276. <https://doi.org/10.1111/1540-4781.00189>
- Nakata, I.S.P. (2000). Learning English vocabulary: Do age and background matter? *TESOL Journal*, 9(2), 6-10. <https://doi.org/10.1002/j.1949-3533.2000.b00239>
- Nakata, I.S.P. (2001). *Learning vocabulary in another language*. Cambridge: Cambridge University Press.
- Nakata, I.S.P. (2013). *Learning vocabulary in another language* (2nd ed.). Cambridge: Cambridge University Press.
- Neale, B., Feagans, L., Pallie, C., & Bae, M. (2006). Re-evaluating the effectiveness of adult literacy education. *Pschonomic Bulletin & Review*, 13, 45-52. <https://doi.org/10.3758/bf03193811>
- Oxford Advanced Learner's Dictionary (10th ed.). (2020). Oxford: Oxford University Press.
- Pahala, E. (2009). Age and English vocabulary acquisition. *ELT Journal*, 63(4), 313-322. <https://doi.org/10.1093/elt/cc014>
- Pea, M. & Rasmussen, E. (2002). The effect of age and background on second language acquisition. *Psychological Research*, 66(3), 180-194. <https://doi.org/10.1007/00426-002-0086-5>
- Pierce, P. (1996). Second language acquisition: The role of age and background. *The Modern Language Journal*, 80(4), 478-493. <https://doi.org/10.1111/j.1540-4781.1996.b05468>
- Pridgen, M., & Dörnyei, P. (2020). Dealing with difficult language: The role of age and background. *Bilingualism: Language and Cognition*, 23(3), 652-667. <https://doi.org/10.1017/S1366728919000622>
- R Development Core Team. (2016). *R: A language and environment for statistical computing*. Vienna: R Foundation for Statistical Computing. <https://www.R-project.org/>
- Schmidt, N. (2008). Reading age: The role of age and background. *Language Teaching Research*, 12(3), 329-363. <https://doi.org/10.1177/1362168808089921>
- Saito, Y., Nakata, T., & Deke, R. (2019). The difficult part of learning a second language: The role of age and background. *The Modern Language Journal*, 103(3), 713-720. <https://doi.org/10.1111/dl.12585>
- Tagahara, K., Kida, S., & Hagiwara, Y. (2010). How difficult? The influence of L1 on second language acquisition. *SLA*, 38(3), 412-421. <https://doi.org/10.1016/j.sla.2010.03.015>
- Tokuda, T. (1993). The effect of age and background on second language acquisition. *SLA*, 21(3), 371-380. [https://doi.org/10.1016/0346-251\(93\)90027-e](https://doi.org/10.1016/0346-251(93)90027-e)
- Tokuda, T. (1997). The effect of age and background on second language acquisition. *Second Language Research*, 13(2), 138-163. <https://doi.org/10.1191/026765897672376469>
- Waring, R. (1997). The effect of age and background on second language acquisition: A meta-analysis. *SLA*, 25(2), 261-274. [https://doi.org/10.1016/0346-251\(97\)00013-4](https://doi.org/10.1016/0346-251(97)00013-4)
- Webb, S., & Chang, A.C.S. (2015). How does age affect second language acquisition? *Studies in Second Language Acquisition*, 37(4), 651-675. <https://doi.org/10.1017/S0272263114000606>

- We che, M. B., & Pa ibakh, T. S. (1996). A e i g ec ld g age cab la k ledge: De h e b ead h. *Canadian Modern Language Review*, 53(1), 13-40. h t t p : / / d o i . o r g / 10.3138 / c m l . 53.1.13
- Wilc , A., & Medi a, A. (2013). Effec f e a ac d h l gical cl e i g L2 cab la ac i i i a g ice lea e . *S stem*, 41(4), 1056-1069. h t t p : / / d o i . o r g / 10.1016 / j . e . 2013.10.012
- Zha g, L., & She g, Y. (2009). A fl h e f l e c e f e a icall - el a e d e f d - e L2 cab la ac i i i . *Foreign Language Learning Theor and Practice*, 29(4), 18-23. (i Chi e e)

A b a a

Ha g S i c e l a l e c e e a c h i g e i e E g l i h e a d i g a d l i g i c c e a Q h U i e i i C h i a . O e f h i e e a c h a e a l i e i i c e d e c l a g a g e (L2) a c i i i . A l g h i l i e , h e i a i c l a l i e e e d i L2 c a b l a a c i i i , i e c e c i e f e e d b a c k a d e a c h e a l k H e a l a k e a f d i e e i L2 l e i c a l c e i g a d a i l e i c a l f e a e a c i a e d i h h i c e .

S a a F a g i c e l a P h D d e l i g i c a h U i e i f P i b g h . H i e e a c h i e e l i e i e c l a g a g e a c i i i f h e e c e i e f h e e i c a l l i g i c a d c h l i g i i e l e i a i c l a l i e e e d i e a i i g h e d i f f e e c e b e e e a i e a d - a i e e a k e i h e c e i g e c h a i i l e d .

1. W d k ledge c i f l i l e c e . I h i d , e e e l e l c c e e d i h h e f a c a f f e c i g f - e a i g l i k . I h e e a i d e f h i a e , e e d d l e a i g i e c h a g e a b l i h f - e a i g c e c i .
2. CET-4 i a l a g e - c a l e a i i d e E g l i h f i c i e c e i h e l l - e a b l i h e d e l i a b i l i a d a l i d i a k e b c l l e g e d e i C h i a , h i c h e a e h e i i i g , l i e i g , e a d i g a d a l a i k i l l a e l l a h e i c a b l a a d g a a k l e d g e i E g l i h .
3. I d e a l l , a l l h e d g e d i l L1 f a i l i a i e h l d b e a e d e h a 3 a e e - i c a l e b e a c h a e . B h e a e k e , a l l f h e h i g h L1 f a i l i a i d h l d e c e i e a c e e l l a b e 4 . H e e , c i d e i g h e e i a l d i f f e e c e i h e i l i f e e e i e c e , h i e i e e a i l d i f f i c l e e . W e e d h e e a a i g c e i e a d , a d e a h i l e e e d h a a l e a 90% f h e a e e h i e i e e f e a c h d .
4. T h e i d e i b e a i e l a i g e d e a c h d e h e h e / h e a a d i e d h e i e i . T h e e e e i e d i d e h i i f a i i a l l h e e a d e i a i e . I a e d f h e a c h i g e l , b e c a e e a c h a i c i a c l e e d e e / e i a i e c c e i e l . W e d i d e h i i f a i e a l a e h e i e f a c e f e a l e a .
5. W e a a a l l i l d , a k i g h e e d e h h a d i l a b a c k g d b d i d a e d h e d e i e e e f d . A b e e i e i e i h h e e e a l e d h a e i e e a a e i e a .

A				
A	A. L	ca	a	
Ta	L1	a	a	N b
				N b
			ab	
	L1	a	a	a
	a	a	a	a
R a -H				
	芹菜	6	3	5.92
	萝卜	6	2	6.44
	紫菜	5	2	6.14
	生菜	7	2	5.99
	辣椒	6	2	5.40
	山药	3	1	6.38
	生姜	6	2	5.86
	韭菜	4	1	5.19
	扁豆	6	2	6.21
	菠菜	7	2	5.81
U a -H				
	刀片	5	1	5.37
	家禽	7	2	5.33
	肾脏	6	2	6.55
	走廊	5	1	5.63
	球拍	6	2	6.15
	玉米棒子	3	1	6.00
	天线	7	3	5.68
	流星	6	2	6.06
		4	1	5.27
	榴莲	6	2	5.96
R a -L				
	响葫芦	7	3	2.78
	大扬琴	8	3	2.24
	扁琴	6	2	2.06
	小军	5	1	3.00
		6	2	2.44
	黑管	8	3	2.26
	低音号	4	2	2.60
	横笛	4	1	3.05
	电子管	5	1	2.94
	小	6	2	2.96
U a -L				
	杜松	7	3	2.03
	球	4	2	2.23
	格子呢	5	1	2.99
	岬角	4	1	2.08
	方解石	7	2	2.22
	梨	7	4	2.96
	吊袜带	6	2	2.65
	修道院	8	2	3.09
	响棒	6	1	2.29
	小胡瓜	8	3	3.18

